ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Kinesthetic Quadratics, Factoring & Polynomial Binomial Arrays

CARD 06 / 10 • ALGB-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · **6·Algebra** · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

GEOMETRIC AREA MODEL. MULTIPLY BY DIMENSION.

Quadratic equations model geometric area.
Polynomial tiles model multiplication geometrically:
Area = Length × Width = (x + a)(x + b)
Arranging tiles into a rectangle factors polynomials!

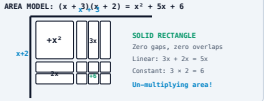
CALIBRATED POLYNOMIAL DATUM

- Unit Datum: +1 Constant Tile = 1U = 7.00 mm pitch
- Linear Rod: +x Tile = 5U × 1U = 35.00 × 7.00 mm
- Quadratic: +x² Tile = 5U × 5U = 35.00 × 35.00 mm
- Connector Spec: 1.85mm base plug & stacking socket

DUAL-STAGE FACTORING LID & L-RAILS

Slide L-rails into lid to set Cartesian origin.
Align tiles against rails to read binomial factors!

AREA MODEL: (x + 3)(x + 2) = x² + 5x + 6



SOLID RECTANGLE
Zero gaps, zero overlaps
Linear: 3x + 2x = 5x
Constant: 3 × 2 = 6
Un-multiplying area!

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Quadratic (+x²)	1x	35.00 × 35.00 mm (5U×5U)	Emerald Green
Quadratic (−x²)	1x	35.00 × 35.00 mm (5U×5U)	Royal Purple
Linear Rod (+x)	5x	35.00 × 7.00 mm (5U×1U)	Emerald Green
Linear Rod (−x)	5x	35.00 × 7.00 mm (5U×1U)	Royal Purple
Constant (+1)	9x	7.00 × 7.00 mm (1U×1U)	Warm Gold
Constant (−1)	9x	7.00 × 7.00 mm (1U×1U)	Crimson Red
Corner L-Rails	2x	105.0 mm Cartesian Axis	Slate Gray
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal
Stage Lid	1x	Dual-Stage Factoring	Matte Charcoal

TOTAL: 30 Math Tiles + 2 L-Rails + Storage Bin + Stage Lid = 34 Parts

DUAL-COLOR SIGN CONVENTION (+ / −)

- Emerald / Gold: Positive mathematical values (+x², +x, +1).
- Royal Purple / Crimson: Negative values (−x², −x, −1).
- Zero-Pair Elimination: Positive cancels negative on sight.
- Matte Texture: Anti-glare finish for document cameras.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Lay the two x² tiles flat on the box floor, slot the 10 x-rods in side bays, and stack the 18 unit cubes in center bays. L-rails snap securely into lid retention. Lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF ALGEBRA
Every degree-n polynomial has exactly n complex roots counted with multiplicity.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 06 OF 10 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING • CARTESIAN L-RAILS • ZERO-PAIR TEXTURE CONTRAST

LESSON GUIDE: FACTORING TRINOMIALS & BINOMIAL ARRAYS

Step-by-Step Hands-On Tutorial: Factoring x² + 5x + 6 = (x+2)(x+3)

Grades 7–11 Pre-Algebra & Algebra II • Hands-On Classroom & Home STEM Studio Series

CARD 06 / 10 • ALGB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/algebra-boxes/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.HSA.SSE.A.2 (Structure)
- CCSS.MATH.CONTENT.HSA.APR.A.1 (Arithmetic)
- TEKS 111.39.c.10.E (Factoring Trinomials)

Grades 7–11 Pre-Algebra & Algebra II

MATHEMATICAL INQUIRY

Can you arrange the linear rods so the rectangle closes completely without any gaps or overlapping edges?

1 STEP 1: SEAT ORIGIN TILE (+x²)

Slide the two L-rails into the lid corner to form the origin.
Place the emerald +x² tile flush into the corner.
Its 35.00 × 35.00 mm area establishes the quadratic core.

2 STEP 2: DISTRIBUTE 5x LINEAR RODS

We need to distribute 5 positive x rods around the +x² tile.
Place 3 rods along the top rail (horizontal dimension).
Place 2 rods along the left rail (vertical dimension).
Notice: 3 + 2 = 5 linear terms accounted for!

3–4 STEPS 3 & 4: COMPLETE RECTANGLE & SOLVE

3. COMPLETE THE RECTANGLE: Fill missing corner with 6 constant (+1) tiles!
4. READ BINOMIAL DIMENSIONS: Horizontal width is (x + 3); vertical length is (x + 2).
Total Area: (x + 2)(x + 3) = x² + 5x + 6!
Calculator Verification: Press Polynomial Solver Icon ([F][EQN] & [F][POLY]) and run [F] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → −2.

5 CORE PRINCIPLE: FACTORING IS UN-MULTIPLYING AREA

Factoring a quadratic is literally finding the length and width of a rectangle! If the tiles cannot form a solid rectangle without gaps, the trinomial is prime. Zero-pairs (+x and −x) let you model differences of squares x² − 4 with effortless physical cancellation!

★ EXTENSION INVESTIGATION: FACTOR: x² + 6x + 8

Arrange one +x² tile, six +x rods, and eight +1 units into a solid rectangle. What are its rail dimensions? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

ALGB-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 34 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California